⌈Def. Define

$$f : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} \frac{1}{p} & \text{if } x \in \mathbb{Q} \text{ with } x = \frac{q}{p}, \text{ irreducible form,} \\ 0 & \text{otherwise.} \end{cases}$$

Then,

1). f is continuous at every irrational numbers, and discontinuous at every rational numbers.

2) f is Riemann-integrable.

Proof. observe that for each $n \in \mathbb{N}$, the set

$$S_n = \{x \in [0,1] \mid f(x) > \tfrac{1}{n}\} = f^{-1}[(\tfrac{1}{n}, \infty)]$$

is finite set. Let $x_0 \in [0,1]$ be an irrational number.

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t $\frac{1}{N} < \varepsilon$. Since the S_N is finite, we can take $\delta > 0$ s.t $\underset{\min\{|x_0 - y| \mid y \in S_N\}}{\underset{\shortparallel}{}}$

$$x \in (x_0 - \delta, x_0 + \delta) \Rightarrow |f(x_0) - f(x)| = f(x) \leq \frac{1}{N} < \varepsilon.$$

So, f is continuous at any irrational. And, for rational $\frac{q}{p} \in [0,1]$,

for any $\delta > 0$, there is an irrational $x \in (\frac{q}{p} - \delta, \frac{q}{p} + \delta)$ s.t $|\frac{1}{p} - f(x)| = \frac{1}{p} > \frac{1}{2p}$.

So it is not continuous at rational. ⌋ 1).

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t $\frac{1}{N} < \varepsilon$.

Denote $S_N = \{x_1, x_2, \cdots, x_k\}$. And, for $0 < d < \min\{|x_i - x_j| \mid 1 \leq i < j \leq k\}$,

Now, construct a partition as

$$P = \{x_1 - \tfrac{d}{2}, x_1 + \tfrac{d}{2}, x_2 - \tfrac{d}{2}, \cdots, x_k - \tfrac{d}{2}, x_k + \tfrac{d}{2}\} \cap [0,1].$$

Then, for the upper sum $\mathcal{U}(f,P)$ and $\mathcal{L}(f,P)$,

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) = \mathcal{U}(f,P) < 1 \cdot \frac{1}{N} + dk < \varepsilon + dk$$

Since the k is fixed, letting $d \to 0$, we obtained the result.